

# Evaluating Fairness in Human-in-the-Loop Machine Learning

A Theoretical and Computational Analysis

## 1. Core Problem and Motivation

Machine Learning systems are increasingly deployed in high-stakes domains (e.g., lending, hiring, justice), prompting intense scrutiny over their fairness. As outlined by Mehrabi et al. (2021), unfairness in AI is not a singular flaw but arises from a continuous **Feedback Loop** comprising three stages: Data  $\rightarrow$  Algorithm  $\rightarrow$  User.

- **Bias in Data:** Models learn from historical data that inherently contains *Historical Bias* (pre-existing societal inequalities) and *Representation Bias* (unequal sampling of populations). Furthermore, *Measurement Bias* occurs when faulty proxies (e.g., arrest rates) are used instead of true metrics (e.g., crime rates).
- **Bias in Algorithms:** Algorithms inject their own bias by mathematically optimizing for overall accuracy, often sacrificing performance on minority groups. This is further compounded by *Ranking Bias*, where systems prioritize specific outputs.
- **Bias in Users:** Humans interacting with the system introduce *Social Bias* and *Behavioral Bias*, completing the loop and generating new, skewed data for future models.

## 2. Taxonomy of Fairness Definitions

Fairness cannot be captured by a single metric. The implementation evaluates the system across three distinct mathematical paradigms.

### 2.1. Group Fairness (Mathematical Parity)

Group fairness mandates that distinct demographic groups receive comparable treatment on average.

- **Demographic Parity:** The rate of positive predictions must be identical across groups:  $P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$ .
- **Equalized Odds:** The algorithm must exhibit equal True Positive Rates (TPR) and False Positive Rates (FPR) across groups.
- **Equal Opportunity:** A relaxation of Equalized Odds requiring only equal TPR, ensuring “qualified” candidates have equal chances.
- **Treatment Equality:** Requires the ratio of False Negatives to False Positives to remain consistent across demographics.

### 2.2. Individual Fairness

Individual fairness dictates that similar cases should be treated similarly, avoiding broad demographic generalizations.

- **Fairness through Unawareness:** Blinding the model to the sensitive attribute. This usually fails because other features act as proxies.
- **Individual Consistency:** Measured via  $k$ -nearest neighbors, ensuring identical feature vectors receive identical predictions.

- **Counterfactual Fairness:** The prediction should remain unchanged in a “parallel universe” where only the sensitive attribute is flipped.

### 2.3. Subgroup Fairness

An intersectional approach demanding that group fairness constraints hold true across hyper-specific subgroups (e.g., Low-Income Minorities vs. High-Income Majorities).

## 3. Human-in-the-Loop (HITL) Dynamics

In critical applications, a human is kept “in the loop” to override algorithmic errors. The simulation categorizes human intervention into distinct behavioral profiles:

### 3.1. The Theoretically Unbiased Human

An ideal, neutral arbiter capable of *Mitigating Algorithmic Bias* and *Solving Omitted Variable Bias*. Because algorithms only know digitized data, an unbiased human can incorporate unmeasured context, pushing the joint prediction closer to ground truth while improving individual fairness.

### 3.2. The Real-World Biased Human

Real-world human annotators often introduce new biases:

- **Automation & Confirmation Bias:** The human blindly trusts the AI’s output, failing to correct errors and reinforcing the AI’s initial bias.
- **Simpson’s Paradox (Aggregation Bias):** The human misapplies broad population statistics to judge individual cases.
- **Explainable Discrimination:** The human justifies unfair outcomes using “logical” but fundamentally biased rationales.

## 4. Simulation & Implementation Details

A Python-based simulation was implemented to empirically evaluate these concepts.

### 4.1. Synthetic Data Generation

A dataset of 3,000 candidates was generated for a binary classification task (loan approval).

- **Representation Bias:** The minority group ( $A = 0$ ) constitutes only 30% of the data.
- **Historical Bias:** Features like `Income` and `CreditScore` were drawn from normal distributions where the majority group had a statistically higher mean, simulating historical economic advantages.

### 4.2. AI Models and Deferral Mechanism

Two logistic regression models were trained:

1. **Unaware AI:** Trained exclusively on non-sensitive features to test the “Fairness through Unawareness” hypothesis.
2. **Aware AI:** Trained on all features. If its prediction probability falls within an uncertainty zone ( $[0.40, 0.60]$ ), the decision is deferred to the human.

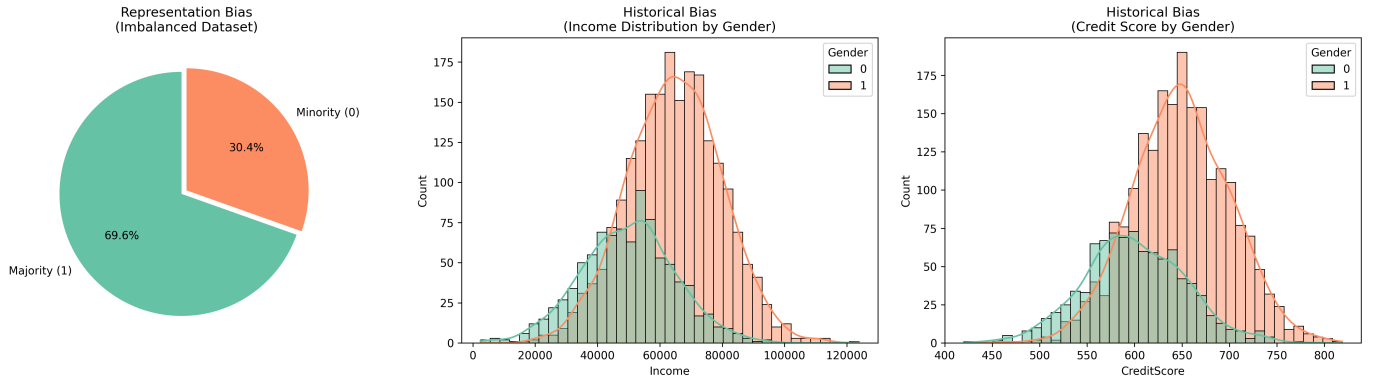


Figure 1: Visualizing Representation Bias (left) and Historical Bias across Income and Credit Score distributions (middle, right).

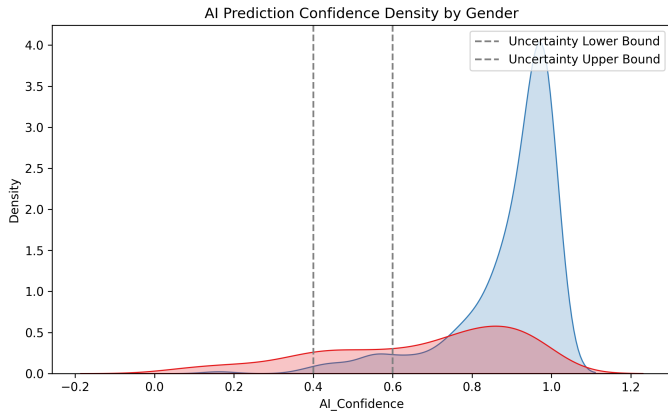


Figure 2: Algorithmic Confidence Density by Gender. The dashed lines denote the uncertainty zone deferred to human annotators.

#### 4.3. Human Scenarios Simulated

For all deferred cases, the simulation applied three human behavioral models:

- **Unbiased Human:** Returns the perfect ground truth.
- **Automation Bias:** Automatically agrees with the AI’s highest probability.
- **Explicit Bias:** Intentionally rejects all minority candidates in the uncertainty zone.

#### 4.4. Visualizing the Bias Pipeline

The simulation generates three critical visual cues that map directly to the theoretical bias pipeline (Data → Algorithm → User):

- **Figure 1 (Data Bias):** The pie chart visually confirms the *Representation Bias* (the 70/30 split), while the KDE distributions prove the embedded *Historical Bias*. It shows that before the AI even trains, the majority group’s features are already skewed higher, setting the stage for subsequent discrimination.
- **Figure 2 (Algorithmic Bias):** The density plot of the AI’s confidence scores demonstrates how the algorithm’s certainty diverges between groups. The dashed lines delineate the threshold where the AI defers to the human,

illustrating the exact subset of data where human biases will be injected into the loop.

- **Figure 3 (User Bias Impact):** The final bar chart synthesizes the entire feedback loop. It serves as visual proof that while an ideal human mitigates bias, a real-world human suffering from Automation or Explicit Bias causes severe degradation across all group fairness constraints (Equalized Odds, Demographic Parity, etc.).

## 5. Experimental Results and Insights

The simulation outputs both terminal-based analytics and graphical charts. Key findings include:

### 5.1. The Illusion of Unawareness

The simulation computationally proves that the “Unaware” AI model still exhibits a significant Demographic Parity penalty. Removing the gender feature is ineffective because **Income** and **CreditScore** act as proxies, allowing the model to reconstruct the historical bias.

### 5.2. Counterfactual Violations

By running a counterfactual pass over the test set, the implementation reveals that a measurable percentage of individuals would have received a *different* outcome had their gender bit been flipped, violating Counterfactual Fairness.

### 5.3. The Price of Fairness (POF)

Adding the Unbiased Human to the loop demonstrates the POF dynamic. In this ideal scenario, the Human-AI team achieves higher overall accuracy while simultaneously reducing the Equal Opportunity gap, proving that mitigating algorithmic bias can sometimes align with performance goals.

### 5.4. The Danger of the Biased Human

The results highlight a stark contrast between theoretical HITL and practical HITL. When Explicit Bias is introduced into the loop, the Equalized Odds difference spikes dramatically compared to the baseline AI.

**Conclusion:** Fairness is not a singular metric but a multi-dimensional constraint. Deploying a “Human-in-the-Loop” is not a panacea for algorithmic bias. If the human is susceptible to automation bias or prejudice, the resulting Human-AI team will actively dismantle the algorithm’s fairness guardrails, requiring strict auditing of human decisions.

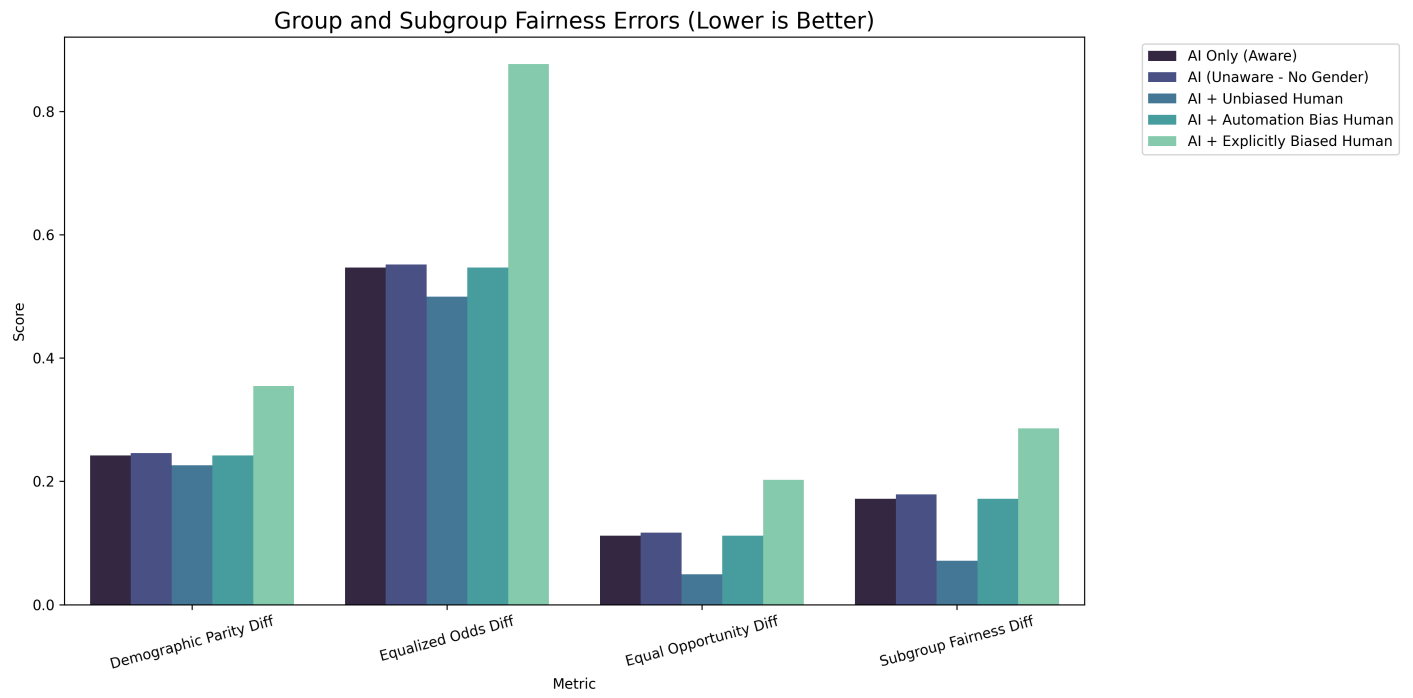


Figure 3: Comparison of Group and Subgroup Fairness Errors across AI-Only and various Human-in-the-Loop configurations.